

Sahithi Reddy Senagapally

Software Engineer (Full-Stack and AI)

Morganville, NJ (Open to Relocation) | +1 (812) 558-4397 | ssenagapally@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Backend engineer with hands-on production experience in Java/Spring Boot microservices, AWS serverless pipelines, and high-throughput data systems. Delivered measurable impact at scale — 83% fraud detection improvement across 1M+ daily transactions at TCS, and a serverless ingestion pipeline that replaced a manual ETL process entirely. Consistently builds systems that are observable, performant, and maintainable under real production load.

TECH STACK

Languages: Java | Python | JavaScript | TypeScript | SQL | C/C++
Frameworks & Libraries: Spring Boot | Node.js | React | Next.js | FastAPI
Architectures & Concepts: Distributed Systems | Microservices | REST API Design | System Design
Databases: MySQL | OracleDB | PostgreSQL | DynamoDB | MongoDB
Cloud & DevOps: AWS (Lambda, S3) | Azure | Docker | GitHub Actions | CI/CD
AI & LLM: LangChain | RAG Pipelines | OpenAI API | Prompt Engineering
Tools & Methods: Linux/Unix | Git | JUnit | TDD | Agile/Scrum

EXPERIENCE

Software Engineer | Indiana University Bloomington | Bloomington, IN Oct 2024 – May 2026

- Resolved concurrent enrollment race conditions for a course management system serving 10,000+ students by implementing MySQL transaction-safe enrollment logic with capacity checks, preventing duplicate enrollments and over-enrollment during peak registration load.
- Built 8+ Node.js REST API endpoints for enrollment, course, and user-management workflows, implementing role-based middleware to enforce access control across student, instructor, and admin tiers.
- Reduced average API response time by 25% by optimizing MySQL queries, adding indexed lookups, and implementing server-side pagination for high-traffic enrollment endpoints.

Software Developer Intern | Isomerous, LLC | Ashburn, VA (Remote) Jun 2025 – Aug 2025

- Replaced a manual spreadsheet-based product inspection workflow by designing and building a serverless ingestion system using API Gateway, Python Lambda, DynamoDB, and S3, enabling real-time submission of structured metadata and base64-encoded images through a single POST endpoint.
- Engineered Lambda processing logic to parse JSON payloads, extract inspection metadata, decode and store product images in product-scoped S3 paths, and persist linked DynamoDB records, eliminating manual data entry and making inspection records traceable and queryable.
- Designed a DynamoDB schema with productId as partition key and ISO 8601 timestamp as sort key, supporting efficient time-range lookups and replacing scattered spreadsheet data with a centralized NoSQL store.

Software Development Engineer | Tata Consultancy Services | Bangalore, India Aug 2023 – Jul 2024

- Improved AML fraud detection true positive rate by 83% by engineering Java/Spring Boot rule engine logic for high-risk transaction patterns, strengthening compliance monitoring across 1M+ daily transactions.
- Cut real-time alert notification delays from 12 minutes to 3 minutes by optimizing Oracle SQL queries, adding targeted indexes, and resolving threading bottlenecks in the transaction processing queue.
- Automated watchlist updates to import regulatory keywords from external sources, improving daily processing speed by 40% through input validation, deduplication, and optimized Oracle database operations.
- Built a real-time monitoring module to ingest AI-generated risk scores, map them to customer profiles, and trigger configurable threshold-based alerts replacing end-of-day batch jobs with instant investigator notifications.

Software Engineer Intern | ADP | Hyderabad, India

July 2022 – Nov 2022

- Built reusable React components for a Report Builder dashboard, implementing filtering, pagination, asynchronous data mapping, and internal REST API integration to support large report datasets in production.
- Collaborated with backend engineers to align API contracts with frontend requirements, independently navigating a large enterprise React codebase to identify and modify components for new dashboard features.

PROJECTS

Distributed Log Processing System

- Built a serverless Azure log-processing pipeline ingesting and parsing 10GB+ of semi-structured logs daily using Blob Storage triggers and Azure Functions, with PostgreSQL indexes on timestamp and severity fields to enable fast incident triage and filtering at scale.

Hoosier Hub - Student Community Platform

- Designed and shipped a full-stack student community platform for 50,000+ IU students, implementing React interfaces, normalized SQL schemas, and scalable REST APIs, with Dockerized services and fully automated Azure CI/CD pipelines to improve reliability and concurrent user access.

EDUCATION

M.S. in Computer Science | Indiana University Bloomington | IN, USA

Aug 2024 – May 2026

B.Tech. in Computer Science | GRIET | Hyderabad, India

Aug 2019 – May 2023